

A Unified Celestial-Body-Agnostic Framework for Variable-Mass Flight Dynamics

From Surface Departure Through Orbit, Atmospheric Entry, Descent, and Landing

Expanded formal mathematical specification

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Abstract

This document formalizes a unified mathematical framework for simulating the motion of variable-mass aerospace vehicles in arbitrary celestial environments. The objective is not to propose a new fundamental law of mechanics, but to provide a common state-space and hybrid-system formulation in which launch, ascent, atmospheric flight, orbital flight, deorbit, entry, descent, landing, staging, deployable devices, ablation, and multi-body events are represented as configurations of one extensible dynamical system. The central body, atmosphere, vehicle, propulsion system, aerodynamic model, thermal model, guidance law, and discrete mission logic are treated as replaceable operators or fields rather than Earth-specific assumptions. The classical Tsiolkovsky rocket equation appears as a limiting analytical case of the framework.

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1 Purpose and Scope

The purpose of this framework is to define a mathematically consistent interface between:

- (i) variable-mass rigid-body dynamics;
- (ii) arbitrary celestial-body gravity, geometry, and rotation;
- (iii) species-resolved atmospheres and general thermodynamic closures;
- (iv) continuum, transitional, and free-molecular aerodynamics;
- (v) propulsion and mass-flow models;
- (vi) aerothermodynamics, thermal response, and ablation;
- (vii) staging, separation, deployable devices, and multi-body dynamics;
- (viii) guidance, navigation, and control;
- (ix) terrain, contact, and touchdown;
- (x) uncertainty, optimization, and verification.

The intended abstraction is

$$\boxed{\dot{\mathbf{X}} = \mathcal{F}_{\kappa}(t, \mathbf{X}, \mathbf{U}; \mathcal{B}, \mathcal{E}, \mathcal{V})} \quad (1)$$

between discrete events, together with jump maps

$$\boxed{g_j(t, \mathbf{X}, \kappa) = 0 \implies (\mathbf{X}^+, \kappa^+) = \mathcal{J}_j(\mathbf{X}^-, \kappa^-).} \quad (2)$$

Here κ is a discrete configuration or mission mode. Equations (1)–(2) define a hybrid dynamical system. When algebraic constraints are also present, the most general formulation is a hybrid differential-algebraic system.

2 Relation to Established Flight-Dynamics Frameworks

The component physics in this document are established. General variable-mass rocket equations have long been derived from momentum balance, including formulations with internal flow, changing inertia, flexibility, and aerodynamic loading [4, 5, 7, 6]. Modern trajectory frameworks likewise combine atmosphere, gravity, propulsion, guidance, multi-vehicle events, and 3-DOF/6-DOF flight.

NASA’s POST2 is particularly close in practical scope: it supports launch, orbital, and entry missions, multiple vehicles with independently defined environments and attracting-body properties, 3-DOF and 6-DOF trajectory segments, user-provided atmosphere/aerodynamic models, events, and Monte Carlo analysis [1, 2]. SimuPy Flight provides an open-source block-composed 6-DOF flight simulation architecture and implements NESC atmospheric verification cases, including motivation from hypersonic re-entry applications [3].

Accordingly, the contribution sought here is a *formal unification and interface specification*, not a claim that the underlying mechanics are novel.

3 Mathematical Objects

Define the complete mission model as

$$\boxed{\mathcal{M} = \{\mathcal{B}, \mathcal{E}, \mathcal{V}, \mathcal{C}, \mathcal{K}\}.} \quad (3)$$

The principal objects are:

$\mathcal{B}$ – celestial body

Gravity potential, surface geometry, orientation, rotation, and optional ephemeris.

$\mathcal{E}$ – environment

Atmospheric/fluid state, radiation, electromagnetic fields, winds, and other spatially varying environmental quantities.

$\mathcal{V}$ – vehicle

Mass properties, geometry, aerodynamics, propulsion, thermal protection, deployables, sensors, actuators, and subsystem states.

$\mathcal{C}$ – guidance and control

Mapping from estimated state, target, time, and mode to control commands.

$\mathcal{K}$ – hybrid configuration

Discrete set of vehicle and mission configurations and associated transition logic.

4 Reference Frames and Conventions

At minimum, define:

- an inertial frame $\mathcal{F}_I$;
- a body-fixed celestial frame $\mathcal{F}_P$;
- a vehicle body frame $\mathcal{F}_B$.

Let $\mathbf{R}_{AB}$ denote the rotation mapping components from frame B to frame A . Vehicle attitude is represented by a unit quaternion $\mathbf{q}_{BI}$ corresponding to $\mathbf{R}_{BI}$, with

$$\|\mathbf{q}_{BI}\| = 1. \quad (4)$$

The baseline formulation uses inertial translational equations. This avoids introducing fictitious Coriolis and centrifugal terms into the primary dynamics; such terms arise only if a rotating frame is selected for integration.

5 Core State

The mandatory 6-DOF core state is

$$\boxed{\mathbf{X}_{\text{core}} = \begin{bmatrix} \mathbf{r}_I \\ \mathbf{v}_I \\ \mathbf{q}_{BI} \\ \boldsymbol{\omega}_B \\ m \end{bmatrix}.} \quad (5)$$

The 3-DOF reduction is

$$\mathbf{X}_{3\text{DOF}} = \begin{bmatrix} \mathbf{r}_I \\ \mathbf{v}_I \\ m \end{bmatrix}. \quad (6)$$

Optional subsystem states are appended by direct sum:

$$\mathbf{X} = \mathbf{X}_{\text{core}} \oplus \bigoplus_{k \in \mathcal{M}_{\text{enabled}}} \mathbf{X}_k. \quad (7)$$

Typical optional states include propulsion, tanks, thermal nodes, TPS recession, structural modes, deployables, landing gear, sensor states, and navigation estimates.

6 Celestial-Body Model

6.1 Gravity

The most general body interface exposes a gravitational potential

$$\Phi(\mathbf{r}_I, t). \quad (8)$$

The gravitational acceleration is

$$\mathbf{g}(\mathbf{r}_I, t) = -\nabla \Phi(\mathbf{r}_I, t). \quad (9)$$

For a spherical isolated body,

$$\Phi = -\frac{\mu}{r}, \quad \mu = GM_P, \quad (10)$$

and therefore

$$\mathbf{g} = -\frac{\mu}{r^3} \mathbf{r}_I. \quad (11)$$

For an oblate body, one possible expansion is

$$\Phi = -\frac{\mu}{r} \left[1 - \sum_{n=2}^N J_n \left(\frac{R_{\text{ref}}}{r} \right)^n P_n(\sin \phi) \right], \quad (12)$$

with extension to tesseral/sectorial harmonics when required. For irregular small bodies, Φ may instead be generated by a polyhedral or mascon model.

6.2 Surface Geometry

Do not require a scalar planetary radius. Define the surface implicitly:

$$S(\mathbf{r}, t) = 0. \quad (13)$$

A signed altitude or clearance may be defined as

$$h = d(\mathbf{r}, S), \quad (14)$$

where d is a signed-distance operator. For a sphere, $h = r - R_P$.

The local outward normal is

$$\hat{\mathbf{n}} = \frac{\nabla S}{\|\nabla S\|}. \quad (15)$$

6.3 Body Rotation

Let the body's inertial orientation be $\mathbf{R}_{IP}(t)$ and its angular velocity be $\boldsymbol{\Omega}_P(t)$. A point fixed to the body has inertial rotational velocity

$$\mathbf{v}_{\text{rot}} = \boldsymbol{\Omega}_P \times \mathbf{r}. \quad (16)$$

7 Environment Model

Define the generic environmental query

$$\boxed{\mathcal{E}(\mathbf{r}, t) \longrightarrow \{\mathbf{g}, h, \hat{\mathbf{n}}, T, p, \rho, \mathbf{v}_f, n_i, \mu_f, k_f, a, \lambda_{\text{mfp}}, \dots\}.} \quad (17)$$

An airless body is represented naturally by $\rho = 0$ and $n_i = 0$; no separate set of equations is required.

7.1 Species-Resolved Atmosphere

For N species, define number densities

$$n_i = n_i(\mathbf{r}, t), \quad i = 1, \dots, N. \quad (18)$$

Total number density is

$$n = \sum_i n_i, \quad (19)$$

mass density is

$$\boxed{\rho = \sum_i n_i m_i,} \quad (20)$$

and mole fraction is

$$x_i = \frac{n_i}{n}. \quad (21)$$

For an ideal mixture,

$$\boxed{p = k_B T \sum_i n_i.} \quad (22)$$

More generally,

$$\boxed{p = \mathcal{P}_{\text{EOS}}(T, \rho, \{x_i\}).} \quad (23)$$

The mixture molar mass and gas constant are

$$\bar{M} = \sum_i x_i M_i, \quad R_m = \frac{R_u}{\bar{M}}. \quad (24)$$

For a calorically perfect mixture,

$$a = \sqrt{\gamma R_m T}, \quad (25)$$

whereas the general thermodynamic definition is

$$\boxed{a^2 = \left(\frac{\partial p}{\partial \rho} \right)_s.} \quad (26)$$

7.2 Fluid Velocity

The atmosphere/fluid module returns its inertial bulk velocity $\mathbf{v}_f$. A useful decomposition is

$$\mathbf{v}_f = \boldsymbol{\Omega}_P \times \mathbf{r} + \mathbf{v}_{\text{wind}}. \quad (27)$$

The vehicle-relative fluid velocity is

$$\boxed{\mathbf{V}_{\text{rel}} = \mathbf{v}_I - \mathbf{v}_f, \quad V_{\text{rel}} = \|\mathbf{V}_{\text{rel}}\|.} \quad (28)$$

8 Dimensionless Flow State

Define

$$\boxed{M = \frac{V_{\text{rel}}}{a}, \quad Re = \frac{\rho V_{\text{rel}} L}{\mu_f}, \quad Kn = \frac{\lambda_{\text{mfp}}}{L}.} \quad (29)$$

Additional dimensionless groups such as Pr , Pe , St , and Da may be supplied as required.

This nondimensional flow state is a key abstraction: aerodynamic regime selection depends primarily on local flow similarity rather than on the name of the celestial body.

9 Aerodynamic and Rarefied-Flow Operator

Define a single aerodynamic operator

$$\boxed{\mathcal{A} : (\mathbf{X}, \mathcal{E}, \kappa) \longrightarrow (\mathbf{F}_A, \mathbf{M}_A, \dot{\mathbf{q}}_{\text{aero}}, \dots).} \quad (30)$$

In continuum flow,

$$q_\infty = \frac{1}{2} \rho V_{\text{rel}}^2, \quad (31)$$

$$\mathbf{F}_A = q_\infty A_{\text{ref}} \mathbf{C}_F, \quad \mathbf{M}_A = q_\infty A_{\text{ref}} L_{\text{ref}} \mathbf{C}_M, \quad (32)$$

where

$$\mathbf{C}_F, \mathbf{C}_M = f(M, Re, Kn, \alpha, \beta, \mathbf{G}, T_w, \{x_i\}, \dots). \quad (33)$$

The implementation may dispatch by regime:

$$\mathcal{A} = \begin{cases} \mathcal{A}_{\text{FM}}, & Kn \gg 1, \\ \mathcal{A}_{\text{trans}}, & Kn \sim 1, \\ \mathcal{A}_{\text{cont}}, & Kn \ll 1. \end{cases} \quad (34)$$

The transition need not be discontinuous; a calibrated bridging law may blend adjacent models.

10 Propulsion and Variable Mass

Define

$$\boxed{\mathcal{P} : (\mathbf{X}, \mathbf{U}, \mathcal{E}, \kappa) \longrightarrow (\mathbf{F}_T, \mathbf{M}_T, \dot{m}, \dot{\mathbf{X}}_{\text{prop}}).} \quad (35)$$

For a conventional rocket nozzle,

$$\boxed{T = \dot{m}_e v_e + (p_e - p_a) A_e,} \quad (36)$$

and

$$\mathbf{F}_T = T \hat{\mathbf{t}}. \quad (37)$$

The mass equation is not fundamentally “rocket-specific”:

$$\boxed{\dot{m} = \sum_k \dot{m}_k}, \quad (38)$$

where the terms may include propellant expulsion, ablation, venting, ingestion, payload release, or other mass transfer.

Care is required in variable-mass mechanics. Momentum carried across the vehicle control surface must be represented either explicitly as momentum-flux terms or consistently inside the thrust/propulsion operator [4, 6, 7].

11 Translational Dynamics

In an inertial frame,

$$\boxed{\dot{\mathbf{r}}_I = \mathbf{v}_I}, \quad (39)$$

and

$$\boxed{m\dot{\mathbf{v}}_I = m\mathbf{g} + \mathbf{F}_T + \mathbf{F}_A + \mathbf{F}_{\text{rad}} + \mathbf{F}_{\text{contact}} + \mathbf{F}_{\text{external}} + \mathbf{F}_{\text{VM}}}. \quad (40)$$

If all mass-flux momentum effects are already included in $\mathbf{F}_T$ and other subsystem forces, then $\mathbf{F}_{\text{VM}} = 0$ by convention.

12 Rotational Dynamics and Mass Properties

Let the body-frame inertia tensor be $\mathbf{I}_B(t)$ and angular velocity $\boldsymbol{\omega}_B$. The angular momentum balance is

$$\frac{d\mathbf{H}}{dt} = \sum \mathbf{M}. \quad (41)$$

For changing inertia,

$$\boxed{\mathbf{I}_B \dot{\boldsymbol{\omega}}_B + \dot{\mathbf{I}}_B \boldsymbol{\omega}_B + \boldsymbol{\omega}_B \times (\mathbf{I}_B \boldsymbol{\omega}_B) = \mathbf{M}_T + \mathbf{M}_A + \mathbf{M}_{\text{contact}} + \mathbf{M}_{\text{external}} + \mathbf{M}_{\text{VM}}}. \quad (42)$$

The center of mass, center of pressure, and inertia tensor may all depend on propellant distribution, staging, deployables, and ablation:

$$\mathbf{r}_{\text{CG}} = \mathbf{r}_{\text{CG}}(\mathbf{X}), \quad \mathbf{I}_B = \mathbf{I}_B(\mathbf{X}). \quad (43)$$

13 Attitude Kinematics

For a unit quaternion,

$$\boxed{\dot{\mathbf{q}}_{BI} = \frac{1}{2} \Omega(\boldsymbol{\omega}_B) \mathbf{q}_{BI}}, \quad (44)$$

where

$$\Omega(\boldsymbol{\omega}_B) = \begin{bmatrix} 0 & -\omega_x & -\omega_y & -\omega_z \\ \omega_x & 0 & \omega_z & -\omega_y \\ \omega_y & -\omega_z & 0 & \omega_x \\ \omega_z & \omega_y & -\omega_x & 0 \end{bmatrix}. \quad (45)$$

14 Aerothermodynamics

Define a thermal operator

$$\boxed{\mathcal{H} : (\mathbf{X}, \mathcal{E}, \kappa) \longrightarrow \dot{\mathbf{X}}_{\text{thermal}}.} \quad (46)$$

For a lumped thermal node j ,

$$m_j c_{p,j} \dot{T}_j = \dot{Q}_{\text{conv},j} + \dot{Q}_{\text{rad},j} - \dot{Q}_{\text{emit},j} - \dot{Q}_{\text{cond},j}, \quad (47)$$

with

$$\dot{Q}_{\text{emit},j} = \epsilon_j \sigma A_j (T_j^4 - T_\infty^4). \quad (48)$$

The heat-flux closure may range from an engineering correlation to a CFD- or database-derived model. The framework constrains the interface, not the fidelity of the closure.

15 Nonequilibrium Chemistry

For high-enthalpy entry, optional chemical states may include

$$\mathbf{X}_{\text{chem}} = [Y_1, \dots, Y_N, T_w, T_e, \dots]^T. \quad (49)$$

A generic species evolution law is

$$\boxed{\dot{Y}_i = \frac{\dot{\omega}_i}{\rho},} \quad (50)$$

where $\dot{\omega}_i$ is the net chemical production rate under the selected kinetics model.

This module permits dissociation, recombination, ionization, vibrational nonequilibrium, and other high-temperature effects without changing the outer equations of motion.

16 Thermal Protection and Ablation

Let

$$\mathbf{X}_{\text{TPS}} = [m_{\text{TPS}}, \delta_{\text{TPS}}, \dots]^T. \quad (51)$$

A generic ablation law is

$$\boxed{\dot{m}_{\text{TPS}} = -\mathcal{B}(\dot{q}, T_w, p, \{x_i\}, \dots).} \quad (52)$$

Geometry evolves through

$$\boxed{\dot{\mathbf{G}} = \mathcal{R}(\mathbf{X}_{\text{thermal}}, \mathbf{X}_{\text{TPS}}).} \quad (53)$$

This creates the feedback chain

$$\text{flow} \rightarrow \text{heating} \rightarrow \text{ablation} \rightarrow \text{geometry/mass} \rightarrow \text{aerodynamics} \rightarrow \text{trajectory}.$$

17 Deployable Devices

A deployable device d has state $\eta_d \in [0, 1]$ and dynamics

$$\dot{\eta}_d = f_d(\eta_d, \mathbf{X}, \mathcal{E}, \mathbf{U}). \quad (54)$$

For a parachute,

$$A_p = A_{p,\text{max}} \eta_p, \quad (55)$$

and a low-order drag model is

$$D_p = \frac{1}{2} \rho V_{\text{rel}}^2 C_{D,p} A_p. \quad (56)$$

Higher-fidelity models may introduce canopy inflation, apparent mass, line dynamics, reefing, or a separate parachute body. The outer interface remains unchanged.

18 Multi-Body Dynamics and Separation

Let the mission contain $N(t)$ active bodies:

$$\boxed{\mathcal{S}(t) = \{\mathbf{X}_1, \dots, \mathbf{X}_{N(t)}\}.} \quad (57)$$

Each body obeys

$$\dot{\mathbf{X}}_i = \mathcal{F}_i(t, \mathbf{X}_i, \mathbf{U}_i, \mathcal{E}, \{\mathbf{X}_j\}_{j \neq i}). \quad (58)$$

A separation event is a jump in system topology:

$$\boxed{\mathbf{X}_{\text{parent}}^- \longrightarrow \{\mathbf{X}_1^+, \mathbf{X}_2^+, \dots\}.} \quad (59)$$

The jump map must conserve linear and angular momentum except for modeled impulses:

$$\mathbf{P}^- + \mathbf{J}_{\text{ext}} = \sum_i \mathbf{P}_i^+. \quad (60)$$

POST2 provides analogous capabilities for initializing additional vehicles from a parent state and transferring angular momentum during events [1].

19 Contact, Terrain, and Landing

Let $h(\mathbf{r})$ denote signed clearance from terrain. A compliant normal-contact model may use

$$F_n = \begin{cases} k\delta + c\dot{\delta}, & \delta = -h > 0, \\ 0, & h \geq 0. \end{cases} \quad (61)$$

A more general unilateral rigid-contact model uses complementarity:

$$\boxed{h \geq 0, \quad F_n \geq 0, \quad hF_n = 0.} \quad (62)$$

Tangential contact may be constrained by Coulomb friction:

$$\|\mathbf{F}_t\| \leq \mu_f F_n. \quad (63)$$

This is one source of algebraic constraints and motivates the DAE extension of the framework.

20 Guidance, Navigation, and Control

20.1 Guidance and Control

Define

$$\boxed{\mathbf{U} = \mathcal{C}(t, \hat{\mathbf{X}}, \mathbf{X}_{\text{target}}, \kappa).} \quad (64)$$

Possible controls include throttle, thrust-vector commands, aerodynamic surfaces, reaction-control jets, bank angle, deployable commands, and landing gear.

20.2 Navigation

Separate the truth state $\mathbf{X}$ from estimated state $\hat{\mathbf{X}}$. Sensors produce

$$\mathbf{y} = \mathcal{H}_s(\mathbf{X}, t) + \boldsymbol{\nu}, \quad (65)$$

and the navigation system evolves

$$\boxed{\dot{\hat{\mathbf{X}}} = \mathcal{N}(\hat{\mathbf{X}}, \mathbf{y}, \mathbf{U}, t).} \quad (66)$$

Guidance should normally consume $\hat{\mathbf{X}}$ rather than perfect truth.

21 Hybrid Mission Modes

Let

$$\mathcal{K} = \{\text{ground, ascent, coast, orbit, deorbit, entry, descent, powered landing, contact}\}. \quad (67)$$

Modes activate or deactivate physical modules rather than replacing the governing mechanics. For example:

$$\kappa = \text{orbit} \Rightarrow \mathcal{A} \approx 0, \quad \mathcal{H} \approx 0,$$

while

$$\kappa = \text{entry} \Rightarrow \mathcal{A}, \mathcal{H}, \mathcal{B} \text{ active.}$$

Typical event functions include

$$g_{\text{burnout}} = m_{\text{prop}}, \quad (68)$$

$$g_{\text{entry}} = h - h_{\text{EI}}, \quad (69)$$

$$g_{\text{chute}} = M - M_{\text{deploy}}, \quad (70)$$

$$g_{\text{jettison}} = q - q_{\text{jettison}}, \quad (71)$$

$$g_{\text{ignite}} = h - h_{\text{ignite}}, \quad (72)$$

$$g_{\text{touchdown}} = h. \quad (73)$$

22 Unified Hybrid DAE Form

The most general compact formulation is

$$\dot{\mathbf{X}} = \mathcal{F}_{\kappa}(t, \mathbf{X}, \mathbf{U}, \mathcal{B}, \mathcal{E}, \mathcal{V}), \quad (74)$$

$$\mathbf{0} = \mathcal{C}_{\text{alg}}(t, \mathbf{X}, \dot{\mathbf{X}}, \mathbf{U}, \kappa), \quad (75)$$

$$g_j(t, \mathbf{X}, \kappa) = 0 \implies (\mathbf{X}^+, \kappa^+) = \mathcal{J}_j(\mathbf{X}^-, \kappa^-). \quad (76)$$

Equation (74) contains continuous dynamics; (75) contains algebraic constraints; and (76) contains discrete events and topology/configuration changes.

23 Constraints and Optimal Control

Engineering constraints may include

$$q_{\infty} \leq q_{\text{max}}, \quad (77)$$

$$\|\mathbf{a}\|_{\text{proper}} \leq a_{\text{max}}, \quad (78)$$

$$\dot{q}_{\text{heat}} \leq \dot{q}_{\text{max}}, \quad (79)$$

$$T_w \leq T_{\text{max}}, \quad (80)$$

$$\sigma_{\text{struct}} \leq \sigma_{\text{allow}}, \quad (81)$$

$$m_{\text{prop}} \geq 0. \quad (82)$$

A trajectory optimization problem is

$$\boxed{\min_{\mathbf{U}(\cdot), t_f, \boldsymbol{\theta}} J(\mathbf{X}(t_f), \mathbf{U}, \boldsymbol{\theta})} \quad (83)$$

subject to the hybrid dynamics, initial conditions, terminal constraints, path constraints, and event logic.

Examples include maximizing delivered mass, minimizing propellant, minimizing peak heat flux, minimizing landing error, or balancing several objectives.

24 Uncertainty and Monte Carlo Formulation

Let uncertain parameters satisfy

$$\boldsymbol{\theta} \sim P(\boldsymbol{\theta}). \quad (84)$$

They may include atmospheric density, winds, aerodynamic coefficients, thrust, mass properties, sensor biases, terrain, and thermal properties.

A Monte Carlo realization is

$$\mathbf{X}^{(k)}(t) = \mathcal{S} \left(\boldsymbol{\theta}^{(k)} \right). \quad (85)$$

Outputs may be expressed as distributions of landing position, heat load, propellant reserve, structural load, or constraint violation probability.

25 Module Fidelity Hierarchy

Level	Module	Function
0	KIN	Position and velocity kinematics
0	GRAV	Gravity field
0	MASS	Mass bookkeeping
0	FORCE	Force accumulation
0	EVENT	Event/root detection and jump maps
0	BODY	Surface geometry and celestial-body state
0	INT	Numerical integration
1	ATT/ROT	Attitude and rigid-body rotation
1	PROP	Propulsion and mass flow
1	ATM	Atmospheric/fluid state
1	AERO	Aerodynamic force and moment closure
1	GUID	Guidance and control
2	RAREFIED	Transitional/free-molecular flow
2	THERM	Aerothermal and structural thermal response
2	CHEM	Nonequilibrium chemistry
2	ABL	Ablation and recession
2	RAD	Radiative heating/pressure
2	GEOM	Time-varying geometry
3	DECEL	Parachutes, ballutes, drag devices
3	MULTI	Multi-body dynamics and separation
3	TERRAIN	Surface topography
3	CONTACT	Touchdown/contact dynamics
3	GEAR	Landing gear
3	NAV	Sensors and navigation
4	SLOSH	Propellant slosh
4	FLEX	Structural flexibility
4	FSI	Fluid-structure interaction
4	PLUME	Plume/environment interaction
4	DUST	Plume-driven surface erosion/ejecta
4	MHD	Magnetohydrodynamic effects
4	ELECTRIC	Charged-particle/electric interactions

26 Minimal Data Contracts

26.1 Celestial Body

A minimal spherical-body implementation requires

$$\boxed{\mathcal{B}_{\min} = \{M_P, R_P, \boldsymbol{\Omega}_P\}.} \quad (86)$$

A general implementation instead provides Φ , S , and body orientation/rotation functions.

26.2 Environment

A minimal atmospheric query should return

$$\boxed{\mathbf{E} = \{\rho, p, T, \mathbf{v}_f, a, \mu_f, \lambda_{\text{mfp}}, \{x_i\}\}.} \quad (87)$$

Quantities not required by the enabled fidelity modules may be omitted or lazily evaluated.

26.3 Vehicle

A minimal point-mass vehicle requires

$$\boxed{\mathcal{V}_{\min} = \{m_0, \mathcal{P}, A_{\text{ref}}, \mathcal{A}\}.} \quad (88)$$

A 6-DOF vehicle adds

$$\{\mathbf{I}_B(\mathbf{X}), \mathbf{r}_{\text{CG}}(\mathbf{X}), \text{geometry}, \mathbf{C}_F, \mathbf{C}_M\}. \quad (89)$$

27 Reduction to Classical Models

27.1 Tsiolkovsky Limit

Set

$$\rho = 0, \quad \mathbf{g} = 0, \quad \mathbf{F}_{\text{external}} = 0,$$

assume fixed thrust direction and constant effective exhaust velocity v_e . Then

$$m \, dv = -v_e \, dm. \quad (90)$$

Integration gives

$$\boxed{\Delta v = v_e \ln \left(\frac{m_0}{m_f} \right).} \quad (91)$$

Thus the Tsiolkovsky equation is an analytical limiting case of the general variable-mass dynamics.

27.2 Keplerian Limit

Set propulsion and atmosphere to zero and use spherical gravity:

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^3} \mathbf{r}. \quad (92)$$

The system reduces to the two-body problem.

27.3 Ballistic Atmospheric Entry

Set $\mathbf{F}_T = 0$ but retain $\mathbf{F}_A$ and $\mathbf{g}$. The framework becomes a ballistic or lifting-entry model depending on the aerodynamic closure.

27.4 Powered Landing

Retain gravity, atmosphere, contact, guidance, and propulsion. No new fundamental equation is required; powered landing is a different control/configuration regime of the same system.

28 Verification and Validation Strategy

The framework should be developed around analytical and independent reference cases.

ID	Test	Primary subsystem
001	Tsiolkovsky rocket equation	Variable mass / propulsion
002	Kepler orbit	Gravity / integration
003	Vacuum free fall	Gravity
004	Rotating-body surface kinematics	Frames / rotation
005	Constant-density drag	Aerodynamics
006	Ballistic atmospheric entry	Entry dynamics
007	Free-molecular drag benchmark	Rarefied flow
008	Lumped thermal slab	Thermal response
009	Momentum-conserving stage separation	Multi-body events
010	Spring-damper touchdown	Contact
011	Quaternion torque-free rotation	Attitude dynamics
012	Mass-depleting rigid-body benchmark	Variable inertia

SimuPy Flight’s use of NESC 6-DOF atmospheric reference cases is an example of this verification philosophy [3]. POST2 likewise uses modular, multi-fidelity trajectory simulation and is applied to ascent, orbital, and entry problems [1, 2].

29 Conservation and Consistency Checks

Where applicable, numerical implementations should monitor:

- mass accounting;
- quaternion normalization;
- linear momentum across separation events;
- angular momentum across separation events;
- mechanical energy in conservative vacuum problems;
- orbital invariants in two-body test cases;
- positivity of density, pressure, temperature, and species fractions;
- $\sum_i Y_i = 1$ and $\sum_i x_i = 1$;
- non-penetration/contact complementarity;
- dimensional consistency in SI units.

For example,

$$m(t) = m_0 + \int_{t_0}^t \dot{m}(\tau) \, d\tau \quad (93)$$

must agree with the sum of all subsystem masses.

30 Numerical Character of the Problem

The framework can become stiff because its time scales may range from orbital periods to shock chemistry, actuator dynamics, or contact transients. Consequently, no single numerical integrator should be prescribed by the mathematics.

The numerical layer should support:

- adaptive explicit Runge–Kutta methods for non-stiff flight phases;
- implicit/stiff solvers when chemistry or thermal networks demand them;
- accurate root finding for event functions;
- state reinitialization after jump maps;
- constraint stabilization or DAE solvers for rigid contact;
- optional symplectic methods for long conservative orbital propagation.

The physical model and numerical integration policy should remain separate interfaces.

31 Formal Notation, Units, and Admissible State Sets

31.1 Dimensions and Units

All exchanged dimensional quantities shall use coherent SI units. Let $[q]$ denote the physical dimension of quantity q , with base dimensions mass M , length L , time T , temperature Θ , amount of substance N , and electric current I . Required dimensions include

$$[\mathbf{r}] = L, \quad [\mathbf{v}] = LT^{-1}, \quad [\mathbf{a}] = LT^{-2}, \quad (94)$$

$$[m] = M, \quad [\mathbf{F}] = MLT^{-2}, \quad [\mathbf{M}] = ML^2T^{-2}, \quad (95)$$

$$[p] = ML^{-1}T^{-2}, \quad [\rho] = ML^{-3}, \quad [\mu_f] = ML^{-1}T^{-1}, \quad (96)$$

$$[k_f] = MLT^{-3}\Theta^{-1}, \quad [c_p] = L^2T^{-2}\Theta^{-1}, \quad [\dot{q}] = MT^{-3}, \quad (97)$$

$$[\Phi] = L^2T^{-2}, \quad [G] = M^{-1}L^3T^{-2}, \quad [\mu] = L^3T^{-2}. \quad (98)$$

Angles, mass fractions, mole fractions, aerodynamic coefficients, and the nondimensional groups $M, Re, Kn, Pr, Pe, St, Da$ are dimensionless.

31.2 Admissible State Domain

The continuous state belongs to a manifold

$$\mathcal{X} \subset \mathbb{R}^3 \times \mathbb{R}^3 \times S^3 \times \mathbb{R}^3 \times \mathbb{R}_{>0} \times \mathcal{Z}, \quad (99)$$

where $S^3 = \{\mathbf{q} \in \mathbb{R}^4 : \mathbf{q}^T \mathbf{q} = 1\}$. Physical admissibility requires, whenever the corresponding variables are enabled,

$$m > 0, \quad \rho \geq 0, \quad p \geq 0, \quad T > 0, \quad (100)$$

$$0 \leq Y_i \leq 1, \quad \sum_i Y_i = 1, \quad 0 \leq x_i \leq 1, \quad \sum_i x_i = 1, \quad (101)$$

$$\mathbf{I}_B = \mathbf{I}_B^T \succ 0, \quad \delta_{\text{TPS}} \geq 0, \quad m_{\text{prop},k} \geq 0. \quad (102)$$

The inertia condition $\mathbf{I}_B \succ 0$ means that $\mathbf{z}^T \mathbf{I}_B \mathbf{z} > 0$ for every nonzero $\mathbf{z} \in \mathbb{R}^3$.

32 Complete Reference-Frame Formalism

Let $\mathcal{F}_I$ be inertial, $\mathcal{F}_P$ central-body-fixed, $\mathcal{F}_L$ local-level, $\mathcal{F}_B$ vehicle-body-fixed, and $\mathcal{F}_W$ wind/relative-flow axes. Rotation matrices satisfy

$$\mathbf{R}_{AB}\mathbf{R}_{BC} = \mathbf{R}_{AC}, \quad \mathbf{R}_{AB}^{-1} = \mathbf{R}_{BA} = \mathbf{R}_{AB}^T. \quad (103)$$

For any geometric vector $\mathbf{a}$,

$$[\mathbf{a}]_A = \mathbf{R}_{AB}[\mathbf{a}]_B. \quad (104)$$

The transport theorem between a frame B rotating with angular velocity $\boldsymbol{\omega}_{BI}$ relative to I is

$$\boxed{\left(\frac{d\mathbf{a}}{dt}\right)_I = \left(\frac{d\mathbf{a}}{dt}\right)_B + \boldsymbol{\omega}_{BI} \times \mathbf{a}.} \quad (105)$$

For position relative to a translating origin O ,

$$\mathbf{v}_I = \mathbf{v}_{O/I} + \left(\frac{d\mathbf{r}}{dt}\right)_B + \boldsymbol{\omega}_{BI} \times \mathbf{r}, \quad (106)$$

$$\mathbf{a}_I = \mathbf{a}_{O/I} + \left(\frac{d^2\mathbf{r}}{dt^2}\right)_B + 2\boldsymbol{\omega}_{BI} \times \left(\frac{d\mathbf{r}}{dt}\right)_B \quad (107)$$

$$+ \dot{\boldsymbol{\omega}}_{BI} \times \mathbf{r} + \boldsymbol{\omega}_{BI} \times (\boldsymbol{\omega}_{BI} \times \mathbf{r}). \quad (108)$$

Hence, if propagation is performed directly in $\mathcal{F}_P$,

$$\boxed{\mathbf{a}_P = \frac{1}{m} \sum \mathbf{F}_P + \mathbf{g}_P - 2\boldsymbol{\Omega}_P \times \mathbf{v}_P - \dot{\boldsymbol{\Omega}}_P \times \mathbf{r}_P - \boldsymbol{\Omega}_P \times (\boldsymbol{\Omega}_P \times \mathbf{r}_P).} \quad (109)$$

This fixes the sign and frame conventions for all rotating-frame terms.

The local-level frame may be constructed from the outward surface normal $\hat{\mathbf{n}}$ and a chosen east-like tangent $\hat{\mathbf{e}}$:

$$\hat{\mathbf{d}} = -\hat{\mathbf{n}}, \quad \hat{\mathbf{n}}_L = \frac{\hat{\mathbf{d}} \times \hat{\mathbf{e}}}{\|\hat{\mathbf{d}} \times \hat{\mathbf{e}}\|}, \quad (110)$$

with appropriate fallback near coordinate singularities. Wind axes are defined by

$$\hat{\mathbf{x}}_W = \frac{\mathbf{V}_{\text{rel}}}{V_{\text{rel}}}, \quad (111)$$

and two orthogonal transverse directions chosen consistently with aerodynamic sign conventions.

33 Control-Volume Derivation of General Variable-Mass Dynamics

33.1 Mass Balance

Let $V(t)$ be a control volume attached to the vehicle, with boundary $S(t)$ moving at local velocity $\mathbf{v}_S$. Let fluid/material velocity be $\mathbf{v}$, density ρ_m , and outward normal $\mathbf{n}$. Reynolds transport gives

$$\frac{d}{dt} \int_{V(t)} \rho_m dV = - \int_{S(t)} \rho_m (\mathbf{v} - \mathbf{v}_S) \cdot \mathbf{n} dA. \quad (112)$$

Define outward mass-flow density

$$\dot{m}'' = \rho_m (\mathbf{v} - \mathbf{v}_S) \cdot \mathbf{n}, \quad (113)$$

positive for outflow. Then

$$\dot{m} = - \int_S \dot{m}'' dA. \quad (114)$$

For discrete ports k with positive outward flow $\dot{m}_k > 0$,

$$\dot{m} = - \sum_{k \in \text{out}} \dot{m}_k + \sum_{k \in \text{in}} \dot{m}_k. \quad (115)$$

33.2 Linear Momentum Balance

The inertial momentum balance is

$$\sum \mathbf{F}_{\text{ext}} = \frac{d}{dt} \int_V \rho_m \mathbf{v} dV + \int_S \rho_m \mathbf{v} [(\mathbf{v} - \mathbf{v}_S) \cdot \mathbf{n}] dA. \quad (116)$$

For a compact vehicle translating at center-of-mass velocity $\mathbf{v}_C$, decompose port velocity as

$$\mathbf{v}_k = \mathbf{v}_C + \boldsymbol{\omega} \times \mathbf{r}_k + \mathbf{u}_k, \quad (117)$$

where $\mathbf{u}_k$ is velocity relative to the local vehicle boundary. The translational equation becomes

$$m \dot{\mathbf{v}}_C = \sum \mathbf{F}_{\text{ext}} - \sum_k s_k \dot{m}_k \mathbf{u}_k - \sum_k s_k \dot{m}_k (\boldsymbol{\omega} \times \mathbf{r}_k) + \mathbf{F}_{\text{int,corr}}, \quad (118)$$

where $s_k = +1$ for outward flow and $s_k = -1$ for inward flow. $\mathbf{F}_{\text{int,corr}}$ collects nonuniform internal-momentum terms if the vehicle cannot be represented by lumped rigid mass plus localized ports. In the common point-port rocket approximation with port at the center of mass and no unresolved internal momentum,

$$m \dot{\mathbf{v}}_C = \sum \mathbf{F}_{\text{ext}} - \sum_k s_k \dot{m}_k \mathbf{u}_k. \quad (119)$$

For a single exhaust stream, $\mathbf{u}_e = -v_e \hat{\mathbf{t}}$ and outward $\dot{m}_e > 0$, giving

$$\mathbf{F}_{\text{mom}} = \dot{m}_e v_e \hat{\mathbf{t}}. \quad (120)$$

Pressure stresses across a nozzle control surface add

$$\mathbf{F}_p = \int_{A_e} (p - p_a) \mathbf{n}_T dA, \quad (121)$$

which reduces for uniform axisymmetric exit conditions to

$$\mathbf{F}_p = (p_e - p_a) A_e \hat{\mathbf{t}}. \quad (122)$$

Thus

$$\mathbf{F}_T = [\dot{m}_e v_e + (p_e - p_a) A_e] \hat{\mathbf{t}} \quad (123)$$

is a specialization of the control-volume momentum balance, not an independent law.

33.3 Angular Momentum Balance

About the instantaneous center of mass C ,

$$\sum \mathbf{M}_C = \left. \frac{d\mathbf{H}_C}{dt} \right|_I + \int_S \mathbf{r} \times \rho_m \mathbf{v} [(\mathbf{v} - \mathbf{v}_S) \cdot \mathbf{n}] dA. \quad (124)$$

For discrete ports, the mass-flow torque is

$$\mathbf{M}_{\text{flux}} = - \sum_k s_k \dot{m}_k \mathbf{r}_k \times \mathbf{u}_k - \sum_k s_k \dot{m}_k \mathbf{r}_k \times (\boldsymbol{\omega} \times \mathbf{r}_k), \quad (125)$$

subject to the same rigid-lumped approximation. Consequently the rigid-body rotational equation is

$$\mathbf{I}\dot{\boldsymbol{\omega}} + \dot{\mathbf{I}}\boldsymbol{\omega} + \boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\omega}) = \mathbf{M}_{\text{ext}} + \mathbf{M}_{\text{flux}} + \mathbf{M}_{\text{int,corr}}. \quad (126)$$

This explicitly defines the previously symbolic $\mathbf{F}_{\text{VM}}$ and $\mathbf{M}_{\text{VM}}$ conventions.

34 General Multi-Body Gravitation and External Fields

For N_c attracting bodies with inertial positions $\mathbf{r}_j(t)$ and gravitational parameters μ_j ,

$$\mathbf{g}(\mathbf{r}, t) = - \sum_{j=1}^{N_c} \mu_j \frac{\mathbf{r} - \mathbf{r}_j}{\|\mathbf{r} - \mathbf{r}_j\|^3} + \mathbf{g}_{\text{harmonics}} + \mathbf{g}_{\text{small-body}}. \quad (127)$$

When one central body's non-spherical field is resolved by potential Φ_P , the total potential may be written

$$\Phi_{\text{total}}(\mathbf{r}, t) = \Phi_P(\mathbf{r}, t) - \sum_{j \neq P} \frac{\mu_j}{\|\mathbf{r} - \mathbf{r}_j\|}. \quad (128)$$

Solar or stellar radiation pressure can be expressed as

$$\mathbf{F}_{\text{SRP}} = P_{\text{rad}}(R_*) C_R A_{\text{proj}} \hat{\mathbf{s}}, \quad P_{\text{rad}}(R_*) = \frac{L_*}{4\pi c R_*^2}, \quad (129)$$

with shadow factor $\chi \in [0, 1]$ included multiplicatively when occultation occurs.

For a vehicle with net charge Q and current distribution represented by effective charge velocity $\mathbf{v}_q$, an optional electromagnetic force closure is

$$\mathbf{F}_{\text{EM}} = Q (\mathbf{E} + \mathbf{v}_q \times \mathbf{B}), \quad (130)$$

with distributed Lorentz forces integrated when a lumped representation is insufficient.

35 Atmospheric Structure and Thermodynamic Closure

35.1 Hydrostatic and Species Transport Closure

A static 1-D atmosphere obeys

$$\frac{dp}{dh} = -\rho g_n(h), \quad (131)$$

where $g_n = -\mathbf{g} \cdot \hat{\mathbf{n}}$. Under an ideal mixture,

$$\frac{d \ln p}{dh} = -\frac{\bar{M}(h) g_n(h)}{R_u T(h)}. \quad (132)$$

For constant T , $\bar{M}$, and g_n ,

$$\boxed{p(h) = p_0 \exp\left(-\frac{h - h_0}{H}\right), \quad H = \frac{R_u T}{\bar{M} g_n}.} \quad (133)$$

A generic species conservation equation for a time-dependent atmosphere is

$$\boxed{\frac{\partial n_i}{\partial t} + \nabla \cdot (n_i \mathbf{v}_f + \mathbf{J}_i) = P_i - L_i,} \quad (134)$$

where $\mathbf{J}_i$ is diffusive number flux and P_i, L_i are chemical/photochemical production and loss. The flight framework may query a prescribed atmospheric model rather than solve these PDEs, but this equation defines what such a model represents.

35.2 Mixture Thermodynamics

Mass fraction and mole fraction are related by

$$Y_i = \frac{x_i M_i}{\bar{M}}, \quad x_i = \frac{Y_i / M_i}{\sum_j Y_j / M_j}. \quad (135)$$

For ideal mixtures,

$$h(T, \mathbf{Y}) = \sum_i Y_i h_i(T), \quad (136)$$

$$u(T, \mathbf{Y}) = \sum_i Y_i u_i(T), \quad (137)$$

$$c_p = \left(\frac{\partial h}{\partial T}\right)_p, \quad (138)$$

$$c_v = \left(\frac{\partial u}{\partial T}\right)_v, \quad (139)$$

$$\gamma = \frac{c_p}{c_v}. \quad (140)$$

A calorically imperfect species may use polynomial heat capacities

$$c_{p,i}(T) = R_i \sum_{k=0}^K a_{ik} T^k, \quad (141)$$

with enthalpy obtained by integration.

35.3 Transport Properties

For dilute gas species with pure viscosities $\mu_i(T)$, one mixture closure is Wilke's rule

$$\boxed{\mu_{\text{mix}} = \sum_i \frac{x_i \mu_i}{\sum_j x_j \phi_{ij}},} \quad (142)$$

where

$$\phi_{ij} = \frac{\left[1 + \sqrt{\mu_i / \mu_j} (M_j / M_i)^{1/4}\right]^2}{\sqrt{8(1 + M_i / M_j)}}. \quad (143)$$

An analogous chosen mixture rule supplies k_f . The framework requires that (μ_f, k_f) be positive and continuous over the atmosphere model's declared domain.

35.4 Mean Free Path and Collision Model

For a single hard-sphere gas,

$$\lambda_{\text{mfp}} = \frac{1}{\sqrt{2} n \sigma_c} = \frac{k_B T}{\sqrt{2} p \sigma_c}. \quad (144)$$

For a mixture, define an effective collision frequency

$$\nu_i = \sum_j n_j \langle \sigma_{ij} g_{ij} \rangle, \quad (145)$$

and species mean free path

$$\lambda_i = \frac{\bar{c}_i}{\nu_i}. \quad (146)$$

The aerodynamic closure may use a selected mixture measure such as $\lambda_{\text{mix}} = \sum_i x_i \lambda_i$ or a directly calibrated effective collision model.

36 Complete Nondimensional Flow Set and Regime Maps

In addition to M , Re , and Kn , define

$$Pr = \frac{\mu_f c_p}{k_f}, \quad (147)$$

$$Pe = Re Pr = \frac{\rho V L c_p}{k_f}, \quad (148)$$

$$St = \frac{\dot{q}}{\rho V c_p (T_r - T_w)}, \quad (149)$$

$$Da = \frac{\tau_{\text{flow}}}{\tau_{\text{chem}}}, \quad \tau_{\text{flow}} = \frac{L}{V}. \quad (150)$$

For compressible reacting entry, a useful local state descriptor is therefore

$$\mathbf{\Pi}_{\text{flow}} = (M, Re, Kn, Pr, Pe, St, Da, \alpha, \beta, \mathbf{Y}, T_w/T_0). \quad (151)$$

36.1 Smooth Regime Transition

To avoid numerical discontinuity between continuum and rarefied models, define a smooth weight on log-Knudsen number. For thresholds $Kn_1 < Kn_2$,

$$\xi = \text{clip} \left(\frac{\log Kn - \log Kn_1}{\log Kn_2 - \log Kn_1}, 0, 1 \right), \quad (152)$$

with cubic smoothstep

$$s(\xi) = 3\xi^2 - 2\xi^3. \quad (153)$$

Then

$$\mathbf{F}_A = (1 - s)\mathbf{F}_{\text{cont}} + s\mathbf{F}_{\text{rare}}, \quad \mathbf{M}_A = (1 - s)\mathbf{M}_{\text{cont}} + s\mathbf{M}_{\text{rare}}. \quad (154)$$

Higher-order or physics-calibrated bridges may replace s , but the implementation must declare its thresholds and continuity class.

37 Aerothermochemical Closure

37.1 Finite-Rate Chemistry

Let reaction r have stoichiometry

$$\sum_i \nu'_{ir} X_i \rightleftharpoons \sum_i \nu''_{ir} X_i, \quad (155)$$

with net coefficient $\nu_{ir} = \nu''_{ir} - \nu'_{ir}$. The molar progress rate is

$$\dot{\xi}_r = k_{f,r} \prod_i C_i^{\nu'_{ir}} - k_{b,r} \prod_i C_i^{\nu''_{ir}}, \quad (156)$$

where C_i is molar concentration. Species mass production is

$$\dot{\omega}_i = M_i \sum_r \nu_{ir} \dot{\xi}_r. \quad (157)$$

Arrhenius forward rates may be written

$$k_{f,r} = A_r T^{\beta_r} \exp\left(-\frac{E_{a,r}}{R_u T}\right). \quad (158)$$

The homogeneous reactor form used in a lumped shock-layer surrogate is

$$\dot{Y}_i = \frac{\dot{\omega}_i}{\rho}, \quad \sum_i \dot{Y}_i = 0. \quad (159)$$

37.2 Multi-Temperature Energy Exchange

If translational-rotational, vibrational, and electron temperatures are separated, let T_{tr}, T_v, T_e be states. A generic energy partition is

$$C_{tr} \dot{T}_{tr} = \dot{Q}_{tr} - Q_{tr \leftrightarrow v} - Q_{tr \leftrightarrow e}, \quad (160)$$

$$C_v \dot{T}_v = \dot{Q}_v + Q_{tr \leftrightarrow v} - Q_{v \leftrightarrow e}, \quad (161)$$

$$C_e \dot{T}_e = \dot{Q}_e + Q_{tr \leftrightarrow e} + Q_{v \leftrightarrow e}, \quad (162)$$

where the exchange terms satisfy pairwise energy conservation.

37.3 Convective and Radiative Heating

A generic surface heat flux is decomposed as

$$\dot{q}_{\text{net}} = \dot{q}_{\text{conv}} + \dot{q}_{\text{rad, in}} - \epsilon \sigma (T_w^4 - T_{\text{env}}^4) - \dot{q}_{\text{conduction}} - \dot{q}_{\text{ablation}}. \quad (163)$$

An engineering stagnation-point correlation may be represented by

$$\dot{q}_{\text{conv, s}} = K_s \sqrt{\frac{\rho_e}{R_n}} V_e^3, \quad (164)$$

where K_s is gas/model dependent and therefore belongs to the declared closure, not the universal constants of the framework.

A gray volumetric radiative-transfer model is governed by

$$\frac{dI_\lambda}{ds} = \kappa_\lambda (B_\lambda - I_\lambda), \quad (165)$$

with incident surface radiative flux

$$\dot{q}_{\text{rad, in}} = \int_0^\infty \int_{\Omega^-} I_\lambda |\hat{\mathbf{s}} \cdot \hat{\mathbf{n}}| d\Omega d\lambda. \quad (166)$$

A lower-fidelity implementation may replace this by a database/correlation returning the same $\dot{q}_{\text{rad, in}}$ interface.

38 Ablation, Recession, and Shape Evolution

At an ablating surface, an energy-balance closure may be written

$$\dot{q}_{\text{conv}} + \dot{q}_{\text{rad,in}} = \epsilon \sigma T_w^4 + k_s \left. \frac{\partial T}{\partial n} \right|_w + \dot{m}_{\text{abl}}'' h_{\text{abl}}. \quad (167)$$

Hence, for prescribed effective heat of ablation h_{abl} ,

$$\dot{m}_{\text{abl}}'' = \max \left(0, \frac{\dot{q}_{\text{in}} - \dot{q}_{\text{rerad}} - \dot{q}_{\text{cond}}}{h_{\text{abl}}} \right). \quad (168)$$

Surface recession is

$$\dot{\delta}_{\text{rec}} = \frac{\dot{m}_{\text{abl}}''}{\rho_{\text{TPS}}}. \quad (169)$$

A level-set representation $\psi(\mathbf{x}, t) = 0$ of the vehicle surface evolves as

$$\frac{\partial \psi}{\partial t} + V_n \|\nabla \psi\| = 0, \quad V_n = -\dot{\delta}_{\text{rec}}, \quad (170)$$

providing a formal geometry-evolution law when shape change is resolved spatially.

39 Flexible Structure, Slosh, and Fluid-Structure Coupling

39.1 Flexible Modes

With generalized elastic coordinates $\boldsymbol{\eta}$,

$$\mathbf{M}_s \ddot{\boldsymbol{\eta}} + \mathbf{C}_s \dot{\boldsymbol{\eta}} + \mathbf{K}_s \boldsymbol{\eta} = \mathbf{Q}_{\text{aero}} + \mathbf{Q}_{\text{prop}} + \mathbf{Q}_{\text{contact}}. \quad (171)$$

Modal truncation gives diagonal $\mathbf{M}_s, \mathbf{K}_s$ under mass-normalized modes, while coupling terms may be retained for large deformation or moving mass.

39.2 Propellant Slosh

A low-order pendulum slosh mode j may be represented by

$$\ddot{\theta}_j + 2\zeta_j \omega_j \dot{\theta}_j + \omega_j^2 \theta_j = -\frac{1}{\ell_j} \mathbf{a}_B \cdot \hat{\mathbf{t}}_j, \quad (172)$$

with slosh force and moment returned to the rigid-body equations. A higher-fidelity solver may replace this surrogate by incompressible free-surface flow while retaining the same resultant force/moment interface.

39.3 Fluid-Structure Interaction

At a fluid-solid interface Γ_{fs} , kinematic and traction continuity require

$$\mathbf{v}_f = \dot{\mathbf{u}}_s, \quad \boldsymbol{\sigma}_f \mathbf{n} = \boldsymbol{\sigma}_s \mathbf{n} \quad \text{on } \Gamma_{fs}. \quad (173)$$

These conditions define the coupling even when fluid and structural PDE solvers are external to the flight integrator.

40 Plume, Surface Interaction, Dust, and MHD Closures

40.1 Plume Forces and Surface Erosion

Let plume stress on a surface be $\mathbf{t}_p = \sigma_p \mathbf{n}$. Resultant plume force and moment are

$$\boxed{\mathbf{F}_{\text{plume}} = \int_{A_p} \mathbf{t}_p dA, \quad \mathbf{M}_{\text{plume}} = \int_{A_p} \mathbf{r} \times \mathbf{t}_p dA.} \quad (174)$$

A generic erosion law is

$$\boxed{\dot{m}''_{\text{erosion}} = K_e [\max(\tau_w - \tau_c, 0)]^{n_e},} \quad (175)$$

where τ_w is plume-induced wall shear and τ_c a threshold. Ejecta may then be represented by particle equations

$$m_p \dot{\mathbf{v}}_p = m_p \mathbf{g} + \mathbf{F}_{D,p} + \mathbf{F}_{\text{other},p}. \quad (176)$$

40.2 Magnetohydrodynamics

For a conducting entry plasma, an MHD closure can add the Lorentz body force

$$\boxed{\mathbf{f}_L = \mathbf{J} \times \mathbf{B},} \quad (177)$$

with Ohm's law, in one common resistive form,

$$\mathbf{J} = \sigma_e (\mathbf{E} + \mathbf{v} \times \mathbf{B}), \quad (178)$$

and Joule heating

$$\dot{q}_J = \mathbf{J} \cdot \mathbf{E}. \quad (179)$$

Maxwell/MHD field evolution may remain external; the vehicle/environment coupling requires resultant forces, moments, and heat addition.

41 Actuator Dynamics and Control Constraints

Controls commanded by guidance need not be achieved instantaneously. Let actuator state $\mathbf{x}_a$ satisfy

$$\boxed{\dot{\mathbf{x}}_a = f_a(\mathbf{x}_a, \mathbf{u}_c), \quad \mathbf{u} = \text{sat}(h_a(\mathbf{x}_a), \mathbf{u}_{\min}, \mathbf{u}_{\max}).} \quad (180)$$

A first-order actuator is

$$\tau_a \dot{u} + u = u_c, \quad (181)$$

subject to rate limits

$$\dot{u}_{\min} \leq \dot{u} \leq \dot{u}_{\max}. \quad (182)$$

Thrust-vector, throttle, flap, reaction-control, parachute, and landing-gear commands are therefore represented as physical states when their dynamics matter.

42 Navigation, Estimation, and Stochastic Dynamics

A continuous stochastic truth model may be written

$$d\mathbf{X} = f(\mathbf{X}, \mathbf{U}, t)dt + \mathbf{G}_w(\mathbf{X}, t)d\mathbf{W}_t, \quad (183)$$

where $\mathbf{W}_t$ is a Wiener process. Discrete measurements at t_k satisfy

$$\boxed{\mathbf{y}_k = h_k(\mathbf{X}(t_k)) + \mathbf{v}_k, \quad \mathbf{v}_k \sim \mathcal{N}(0, \mathbf{R}_k).} \quad (184)$$

For an extended Kalman filter, covariance evolves between measurements as

$$\dot{\mathbf{P}} = \mathbf{A}\mathbf{P} + \mathbf{P}\mathbf{A}^T + \mathbf{G}_w\mathbf{Q}\mathbf{G}_w^T, \quad \mathbf{A} = \left. \frac{\partial f}{\partial \mathbf{X}} \right|_{\hat{\mathbf{X}}}, \quad (185)$$

and the measurement update is

$$\mathbf{K}_k = \mathbf{P}^- \mathbf{H}_k^T (\mathbf{H}_k \mathbf{P}^- \mathbf{H}_k^T + \mathbf{R}_k)^{-1}, \quad (186)$$

$$\hat{\mathbf{X}}^+ = \hat{\mathbf{X}}^- + \mathbf{K}_k (\mathbf{y}_k - h_k(\hat{\mathbf{X}}^-)), \quad (187)$$

$$\mathbf{P}^+ = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k) \mathbf{P}^-. \quad (188)$$

Other estimators may replace this closure without changing the truth-model interface.

43 Hybrid Event Semantics and Topology Changes

43.1 Guard Sets and Directionality

For mode κ , event j has guard

$$G_j = \{(t, \mathbf{X}) : g_j(t, \mathbf{X}, \kappa) = 0\}. \quad (189)$$

An event is eligible only if its declared crossing direction $d_j \in \{-1, 0, +1\}$ is satisfied:

$$\text{sgn}\left(\frac{dg_j}{dt}\right) = d_j \quad \text{when } d_j \neq 0. \quad (190)$$

Root isolation must locate t_e such that

$$|g_j(t_e, \mathbf{X}(t_e))| \leq \varepsilon_g. \quad (191)$$

43.2 Simultaneous Events

Define the active event set at time t_e as

$$\mathcal{A}_e = \{j : |g_j(t_e, \mathbf{X}^-)| \leq \varepsilon_g\}. \quad (192)$$

Each event has integer priority π_j . Events are ordered by decreasing priority; equal-priority events must either commute,

$$\mathcal{J}_i \circ \mathcal{J}_j = \mathcal{J}_j \circ \mathcal{J}_i, \quad (193)$$

or be combined into an explicitly defined composite jump

$$\mathcal{J}_{ij} : \mathbf{X}^- \mapsto \mathbf{X}^+. \quad (194)$$

After each jump, all guards are re-evaluated at the same physical time until no newly enabled zero-time event remains. A maximum jump count N_{Zeno} prevents unresolved Zeno behavior and forces an explicit error if exceeded.

43.3 Topology-Changing State Maps

Let a system state be a labeled collection

$$\mathcal{S}^- = (\mathcal{I}^-, \{\mathbf{X}_i^-\}_{i \in \mathcal{I}^-}). \quad (195)$$

A topology jump is

$$\boxed{\mathcal{J}_j : (\mathcal{I}^-, \{\mathbf{X}_i^-\}) \mapsto (\mathcal{I}^+, \{\mathbf{X}_i^+\})}. \quad (196)$$

For separation of parent p into children a, b , required constraints are

$$m_p^- = m_a^+ + m_b^+ + m_{\text{expelled}}, \quad (197)$$

$$\mathbf{P}_p^- + \mathbf{J}_{\text{sep}} = \mathbf{P}_a^+ + \mathbf{P}_b^+ + \mathbf{P}_{\text{expelled}}, \quad (198)$$

$$\mathbf{H}_p^- + \mathbf{L}_{\text{sep}} = \mathbf{H}_a^+ + \mathbf{H}_b^+ + \mathbf{H}_{\text{expelled}}. \quad (199)$$

Child positions and attitudes are generated by declared geometric transforms from the parent configuration, thereby making vehicle creation deterministic.

44 Differential-Algebraic Constraints and Well-Posedness

Write a mode as an implicit DAE

$$\boxed{\mathbf{F}_\kappa(t, \mathbf{X}, \dot{\mathbf{X}}, \mathbf{U}) = \mathbf{0}.} \quad (200)$$

For a semi-explicit index-1 form,

$$\dot{\mathbf{x}} = f(t, \mathbf{x}, \mathbf{z}, \mathbf{U}), \quad (201)$$

$$\mathbf{0} = g(t, \mathbf{x}, \mathbf{z}, \mathbf{U}). \quad (202)$$

Local solvability for $\mathbf{z}$ is guaranteed when

$$\boxed{\det \left(\frac{\partial g}{\partial \mathbf{z}} \right) \neq 0} \quad (203)$$

by the implicit-function theorem. For a pure ODE mode $\dot{\mathbf{X}} = f(t, \mathbf{X})$, local existence and uniqueness follow if f is continuous in t and locally Lipschitz in $\mathbf{X}$ on the admissible domain.

For unilateral contact represented by complementarity,

$$0 \leq h(\mathbf{X}) \perp \lambda_n \geq 0, \quad (204)$$

where λ_n is normal contact force. Contact-mode solvability requires a nonsingular constrained mass matrix and a consistent active set. At every jump, the post-event state must satisfy the algebraic consistency condition

$$\mathcal{C}_{\text{alg}}(t_e, \mathbf{X}^+, \dot{\mathbf{X}}^+, \mathbf{U}^+, \kappa^+) = 0. \quad (205)$$

The framework declares a model segment *well posed* on domain D if: (i) all enabled closures are defined on D ; (ii) the continuous problem has a unique local solution; (iii) event guards are measurable and root-isolatable; (iv) jump maps produce admissible, algebraically consistent states; and (v) no unresolved accumulation of infinitely many jumps occurs in finite time.

45 Contact with Impacts and Friction

For rigid impact at normal gap $h = 0$ with pre-impact normal relative velocity $v_n^- < 0$, define normal impulse $J_n \geq 0$ and coefficient of restitution $e_n \in [0, 1]$:

$$\boxed{v_n^+ = -e_n v_n^-}. \quad (206)$$

The generalized velocity jump is

$$\mathbf{M}_q(\dot{\mathbf{q}}^+ - \dot{\mathbf{q}}^-) = \mathbf{J}_c^T \boldsymbol{\Lambda}, \quad (207)$$

where $\mathbf{J}_c$ is the contact Jacobian and $\boldsymbol{\Lambda}$ the impulse vector. Coulomb friction satisfies

$$\|\boldsymbol{\Lambda}_t\| \leq \mu_f \Lambda_n, \quad (208)$$

with sticking when the inequality is strict and sliding when equality holds opposite tangential motion.

46 Relativistic Extension

For ordinary planetary flight, Newtonian mechanics is sufficient. If relativistic correction is enabled, use four-position x^μ , four-velocity $u^\mu = dx^\mu/d\tau$, and four-momentum $p^\mu = mu^\mu$. In flat spacetime with applied four-force f^μ ,

$$\boxed{\frac{dp^\mu}{d\tau} = f^\mu, \quad u^\mu u_\mu = -c^2.} \quad (209)$$

For geodesic free fall in curved spacetime,

$$\boxed{\frac{d^2x^\mu}{d\tau^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0.} \quad (210)$$

This module replaces, rather than augments, the Newtonian gravity/kinematics core when its declared applicability threshold is exceeded.

47 Formal Optimal-Control Problem with Hybrid Phases

For phases $k = 1, \dots, N_p$, let state/control be $(\mathbf{X}_k, \mathbf{U}_k)$ over $[t_k^0, t_k^f]$. A general Bolza objective is

$$\boxed{J = \Phi_f(\mathbf{X}_{N_p}(t_f), t_f) + \sum_{k=1}^{N_p} \int_{t_k^0}^{t_k^f} L_k(t, \mathbf{X}_k, \mathbf{U}_k) dt.} \quad (211)$$

Subject to

$$\dot{\mathbf{X}}_k = f_k(t, \mathbf{X}_k, \mathbf{U}_k), \quad (212)$$

$$\mathbf{c}_k(t, \mathbf{X}_k, \mathbf{U}_k) \leq 0, \quad (213)$$

$$\mathbf{b}_k(\mathbf{X}_k(t_k^0), \mathbf{X}_k(t_k^f)) = 0, \quad (214)$$

and phase linkage

$$\mathbf{X}_{k+1}(t_{k+1}^0) = \mathcal{J}_k(\mathbf{X}_k(t_k^f)). \quad (215)$$

For smooth unconstrained arcs, Pontryagin's Hamiltonian is

$$\mathcal{H} = L + \boldsymbol{\lambda}^T f, \quad (216)$$

with necessary conditions

$$\dot{\boldsymbol{\lambda}} = -\frac{\partial \mathcal{H}}{\partial \mathbf{X}}, \quad \mathbf{U}^* = \arg \min_{\mathbf{U} \in \mathcal{U}} \mathcal{H}. \quad (217)$$

Numerical implementation may instead use direct collocation; these equations define the continuous optimum being approximated.

48 Formal Uncertainty Propagation and Sensitivity

Let parameters be $\boldsymbol{\theta}$ and dynamics $\dot{\mathbf{X}} = f(\mathbf{X}, \boldsymbol{\theta}, t)$. First-order state sensitivity

$$\mathbf{S}_\theta = \frac{\partial \mathbf{X}}{\partial \boldsymbol{\theta}} \quad (218)$$

satisfies

$$\boxed{\dot{\mathbf{S}}_\theta = \mathbf{A} \mathbf{S}_\theta + \mathbf{B}_\theta, \quad \mathbf{A} = \frac{\partial f}{\partial \mathbf{X}}, \quad \mathbf{B}_\theta = \frac{\partial f}{\partial \boldsymbol{\theta}}.} \quad (219)$$

Across a differentiable jump $\mathbf{X}^+ = \mathcal{J}(\mathbf{X}^-, \boldsymbol{\theta})$,

$$\mathbf{S}_\theta^+ = \frac{\partial \mathcal{J}}{\partial \mathbf{X}} \mathbf{S}_\theta^- + \frac{\partial \mathcal{J}}{\partial \boldsymbol{\theta}}, \quad (220)$$

with an event-time correction when the guard time depends on parameters. Monte Carlo remains the nonlocal reference method when distributions are strongly nonlinear or multimodal.

49 Verification Metrics and Numerical Acceptance Tolerances

Every reference test declares an exact/reference solution $y_{\text{ref}}(t)$, numerical result $y_h(t)$, and scaling $s_y > 0$. Define normalized maximum error

$$E_{\infty}(y) = \max_t \frac{\|y_h(t) - y_{\text{ref}}(t)\|}{s_y + \|y_{\text{ref}}(t)\|}. \quad (221)$$

For invariants $I(\mathbf{X})$,

$$E_I = \max_t \frac{|I(\mathbf{X}(t)) - I(\mathbf{X}(t_0))|}{I_{\text{scale}}}. \quad (222)$$

For an event time,

$$E_t = \frac{|t_{e,h} - t_{e,\text{ref}}|}{\max(t_{\text{scale}}, |t_{e,\text{ref}}|)}. \quad (223)$$

A baseline double-precision verification profile is proposed as

$$E_{\infty} \leq 10^{-8} \quad \text{for smooth analytical ODE tests,} \quad (224)$$

$$E_I \leq 10^{-9} \quad \text{for short conservative invariant tests,} \quad (225)$$

$$E_t \leq 10^{-9} \quad \text{for isolated analytical event times,} \quad (226)$$

$$\frac{|\sum m_i^+ - m^-|}{m^-} \leq 10^{-12} \quad \text{for exact mass-conserving jumps,} \quad (227)$$

$$\frac{\|\Delta \mathbf{P}\|}{P_{\text{scale}}} \leq 10^{-10} \quad \text{for momentum-conserving separation tests.} \quad (228)$$

These are reference targets, not universal solver tolerances. Each benchmark must publish its own numerical method, step/error tolerance, precision, and expected convergence order. A mesh/step refinement study should verify

$$E(h) \approx Ch^p \quad (229)$$

with observed order

$$p_{\text{obs}} = \frac{\log(E(h_1)/E(h_2))}{\log(h_1/h_2)}. \quad (230)$$

50 Model-Fidelity Declaration and Provenance

Every simulation result shall be associated with a fidelity vector

$$\boldsymbol{\ell} = (\ell_{\text{grav}}, \ell_{\text{atm}}, \ell_{\text{aero}}, \ell_{\text{prop}}, \ell_{\text{therm}}, \ell_{\text{chem}}, \ell_{\text{TPS}}, \ell_{\text{struct}}, \ell_{\text{contact}}, \ell_{\text{GNC}}, \dots), \quad (231)$$

where each ℓ_j is a declared model identifier or ordered fidelity level, not merely a Boolean. Define the model declaration

$$\mathfrak{D} = (\boldsymbol{\ell}, \boldsymbol{\theta}, \mathcal{D}_{\text{source}}, \mathcal{N}, \varepsilon, \mathcal{V}_{\text{code}}), \quad (232)$$

where $\boldsymbol{\theta}$ are model parameters, $\mathcal{D}_{\text{source}}$ source datasets, $\mathcal{N}$ numerical method, ε solver tolerances, and $\mathcal{V}_{\text{code}}$ implementation/version identity. Two simulations are formally reproducible only if their declarations are equivalent to the level required by the comparison.

51 Formal Operator Contracts

For implementation-independent mathematical consistency, each enabled operator must declare a domain D , codomain C , differentiability/continuity class, units, and validity region. Representative contracts are

$$\mathcal{B} : (\mathbf{r}, t) \in D_B \mapsto (\Phi, \mathbf{g}, S, \hat{\mathbf{n}}, \mathbf{R}_{IP}, \mathbf{\Omega}_P), \quad (233)$$

$$\mathcal{E} : (\mathbf{r}, t) \in D_E \mapsto (T, p, \rho, \mathbf{v}_f, \mathbf{x}, \mu_f, k_f, a, \lambda), \quad (234)$$

$$\mathcal{A} : (\mathbf{X}, \mathcal{E}, \kappa) \in D_A \mapsto (\mathbf{F}_A, \mathbf{M}_A, \dot{q}_A), \quad (235)$$

$$\mathcal{P} : (\mathbf{X}, \mathbf{U}, \mathcal{E}, \kappa) \in D_P \mapsto (\mathbf{F}_T, \mathbf{M}_T, \dot{\mathbf{m}}, \dot{\mathbf{X}}_P), \quad (236)$$

$$\mathcal{H} : (\mathbf{X}, \mathcal{E}, \kappa) \in D_H \mapsto \dot{\mathbf{X}}_H, \quad (237)$$

$$\mathcal{C} : (t, \hat{\mathbf{X}}, \mathbf{X}_{\text{target}}, \kappa) \in D_C \mapsto \mathbf{U}_C. \quad (238)$$

If a query lies outside an operator's validity region, the framework must not silently extrapolate unless the model declaration explicitly allows and labels the extrapolation.

52 Complete Coupled State Equation

With all baseline modules enabled, a representative expanded state is

$$\mathbf{X} = [\mathbf{r}, \mathbf{v}, \mathbf{q}, \boldsymbol{\omega}, \mathbf{m}_{\text{tanks}}, \mathbf{x}_{\text{engine}}, \boldsymbol{\eta}, \dot{\boldsymbol{\eta}}, \mathbf{T}_s, \mathbf{Y}, T_v, T_e, \boldsymbol{\delta}_{\text{TPS}}, \boldsymbol{\eta}_{\text{dev}}, \mathbf{x}_a, \hat{\mathbf{X}}, \text{vec } \mathbf{P}]^T. \quad (239)$$

The coupled equations may be summarized as

$$\begin{aligned} \dot{\mathbf{r}} &= \mathbf{v}, \\ m\dot{\mathbf{v}} &= m\mathbf{g} + \mathbf{F}_{\text{aero}} + \mathbf{F}_{\text{prop}} + \mathbf{F}_{\text{rad}} + \mathbf{F}_{\text{EM}} + \mathbf{F}_{\text{contact}} + \mathbf{F}_{\text{flux}}, \\ \dot{\mathbf{q}} &= \frac{1}{2}\Omega(\boldsymbol{\omega})\mathbf{q}, \\ \mathbf{I}\dot{\boldsymbol{\omega}} &= \mathbf{M}_{\text{total}} - \dot{\mathbf{I}}\boldsymbol{\omega} - \boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\omega}), \\ \dot{\mathbf{m}}_{\text{tanks}} &= \mathbf{f}_m, \\ \dot{\mathbf{x}}_{\text{engine}} &= \mathbf{f}_P, \\ \mathbf{M}_s\ddot{\boldsymbol{\eta}} + \mathbf{C}_s\dot{\boldsymbol{\eta}} + \mathbf{K}_s\boldsymbol{\eta} &= \mathbf{Q}_s, \\ \mathbf{C}_T\dot{\mathbf{T}}_s &= \mathbf{Q}_{\text{heat}}(\mathbf{X}, \mathcal{E}), \\ \dot{\mathbf{Y}} &= \boldsymbol{\omega}_{\text{chem}}/\rho, \\ \dot{T}_v &= f_v, \quad \dot{T}_e = f_e, \\ \dot{\boldsymbol{\delta}}_{\text{TPS}} &= \mathbf{f}_{\text{rec}}, \\ \dot{\boldsymbol{\eta}}_{\text{dev}} &= \mathbf{f}_{\text{dev}}, \\ \dot{\mathbf{x}}_a &= \mathbf{f}_a, \\ \dot{\hat{\mathbf{X}}} &= \mathbf{f}_N, \quad \dot{\mathbf{P}} = \mathbf{f}_{\text{cov}}, \\ \mathbf{0} &= \mathcal{C}_{\text{alg}}(t, \mathbf{X}, \hat{\mathbf{X}}, \mathbf{U}, \kappa). \end{aligned} \quad (240)$$

Every right-hand-side closure above has been defined in the present document or as a declared replaceable operator with explicit domain/codomain requirements.

53 Proposed Formal Definition

Definition. A *Celestial-Body-Agnostic Hybrid Flight Dynamics System* is a tuple

$$\mathfrak{F} = (\mathcal{X}, \mathcal{U}, \mathcal{K}, \mathcal{B}, \mathcal{E}, \mathcal{V}, \{\mathcal{F}_\kappa\}_{\kappa \in \mathcal{K}}, \{g_j, \mathcal{J}_j\}_{j \in \mathcal{J}}) \quad (241)$$

such that:

1. $\mathcal{X}$ is a continuous state manifold containing at minimum translational state and mass;
2. $\mathcal{U}$ is the admissible control set;
3. $\mathcal{K}$ is a discrete configuration set;
4. $\mathcal{B}$ supplies gravity, surface geometry, and body kinematics;
5. $\mathcal{E}$ supplies local environmental fields;
6. $\mathcal{V}$ supplies mass properties and force/moment/subsystem closures;
7. $\mathcal{F}_\kappa$ defines the continuous equations in each mode;
8. g_j defines event guards;
9. $\mathcal{J}_j$ defines state/configuration/topology changes at events.

A mission trajectory is then a piecewise-smooth hybrid solution $(\mathbf{X}(t), \kappa(t))$ satisfying the continuous equations, algebraic constraints, event guards, and jump maps over the mission interval.

54 Interpretation

The principal conceptual result is that

surface departure $\rightarrow$ atmospheric ascent $\rightarrow$ spaceflight $\rightarrow$ entry $\rightarrow$ descent $\rightarrow$ surface contact

need not be represented by separate fundamental simulators.

They are regimes of one system:

$$\dot{\mathbf{X}} = \mathcal{F}_\kappa(t, \mathbf{X}, \mathbf{U}; \mathcal{B}, \mathcal{E}, \mathcal{V}). \quad (242)$$

Likewise, “Earth”, “Mars”, “Titan”, an airless moon, or an irregular asteroid are not different equations. They are different implementations or parameterizations of $\mathcal{B}$ and $\mathcal{E}$.

55 Boundary of the Framework

This framework is intended to unify flight mechanics and its principal coupled physical closures. It does not imply that every closure can be reduced to a simple analytic formula. In particular, high-fidelity CFD, DSMC, reacting chemistry, ablation, structural finite-element models, and plume-surface interaction may remain external numerical models or surrogate databases. The framework specifies how their outputs couple into the global state equations.

Likewise, relativity, multi-body celestial mechanics, electromagnetic propulsion, or strongly coupled plasma physics can be added through the same operator structure when a mission requires them, but are not mandatory for the baseline specification.

56 Mathematical Completeness Status

The items previously identified as requiring further formalization are now incorporated explicitly in the preceding and following sections: symbol dimensions and admissible sets; frame transformations; control-volume variable-mass mechanics; thermodynamic and transport closures; hybrid-event precedence; topology-changing jump maps; aerodynamic regime blending; ODE/DAE well-posedness; verification tolerances; and a model-fidelity declaration. The framework remains extensible because particular closures may be replaced by higher-fidelity models, but every module named in the baseline architecture now has a mathematical interface and consistency conditions.

57 Conclusion

The unified framework is most naturally expressed not as a single extended rocket equation, but as a hybrid, variable-mass, multiphysics state-space system. Tsiolkovsky’s equation, Keplerian orbital motion, ballistic entry, lifting entry, powered descent, staging, and touchdown all arise as reductions or configurations of the same general structure.

The central abstraction is:

$$\begin{aligned} \dot{\mathbf{X}} &= \mathcal{F}_\kappa(t, \mathbf{X}, \mathbf{U}, \mathcal{B}, \mathcal{E}, \mathcal{V}), \\ \mathbf{0} &= \mathcal{C}_{\text{alg}}(t, \mathbf{X}, \dot{\mathbf{X}}, \mathbf{U}, \kappa), \\ g_j(t, \mathbf{X}, \kappa) = 0 &\Rightarrow (\mathbf{X}^+, \kappa^+) = \mathcal{J}_j(\mathbf{X}^-, \kappa^-). \end{aligned} \tag{243}$$

This formulation provides a coherent mathematical foundation on which a planet-agnostic, surface-to-space-to-surface simulation architecture can subsequently be designed.

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